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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
3rd SEMESTER B.TECH DEGREE EXAM DEC '2020

MAT 201: PARTIAL DIFFERENTIAL EQUATIONS
AND COMPLEX ANALYSIS.

PART A

① $z = f(x^2 - y^2)$

$$p = f'(x^2 - y^2)(2x), \quad q = f'(x^2 - y^2)(-2y)$$

$$\frac{p}{q} = \frac{2x}{-2y} \Rightarrow \underline{p_y + q_x = 0}$$

② $\frac{\partial^2 z}{\partial x^2} + z = 0$

Annihilary equation is $m^2 + 1 = 0 \Rightarrow m = \pm i$

$$z = c_1 \cos x + c_2 \sin x.$$

$$\Rightarrow z = f(y) \cos x + g(y) \sin x. \rightarrow \textcircled{1}$$

Diff. ① w.r.t 'x',

$$\frac{\partial z}{\partial x} = f(y)(-\sin x) + g(y) \cos x \rightarrow \textcircled{2}$$

$$x=0, \Rightarrow z = e^y$$

$$\textcircled{1} \Rightarrow \boxed{e^y = f(y)}$$

$$x=0 \Rightarrow \frac{\partial z}{\partial x} = 1$$

$$\textcircled{2} \Rightarrow \boxed{1 = g(y)}$$

$$\therefore \underline{z = e^y \cos x + \sin x}$$

(2)

③ Conditions $y(0,t)=0, y(l,t)=0, y(x,0)=0$

$$\frac{\partial y}{\partial t}(x,0) = v_0 \sin^3\left(\frac{\pi x}{l}\right)$$

④ $u(x,t) = (C_1 e^{px} + C_2 e^{-px}) e^{c^2 p^2 t}$

$$u(x,t) = (C_3 \cos px + C_4 \sin px) e^{-c^2 p^2 t}$$

$$u(x,t) = (C_5 x + C_6)$$

⑤ $f(z) = 5z^2 - 12z + 3 + 2i \rightarrow \textcircled{1}$

Put $z = x + iy$

$\textcircled{1} \Rightarrow f(z) = u + iv = 5(x+iy)^2 - 12(x+iy) + 3 + 2i$

$\Rightarrow u = 5x^2 - 5y^2 - 12x + 3, v = 10xy - 12y + 2$

$u(4, -3) = -10$, $v(4, -3) = -82$

⑥ $w = (a+ib)z^2$

Fixed points are given by $z = (a+ib)z^2$

$\Rightarrow z^2(a+ib) - z = 0$

$\Rightarrow z[(a+ib)z - 1] = 0 \Rightarrow z = 0, \frac{1}{a+ib}$

ie $z = 0, \frac{a-ib}{a^2+b^2}$

(3)

$$(7) \oint_C \frac{e^z}{z-2} dz, \quad |z|=3$$

Singularity $z=2$ lies inside $|z|=3$

By Cauchy's Integral Formula,

$$\oint_C \frac{f(z)}{z-z_0} dz = 2\pi i f(z_0).$$

$$\oint_C \frac{e^z}{z-2} dz = 2\pi i \times f(2) \quad \left| \begin{array}{l} f(z) = e^z \\ f(2) = e^2 \end{array} \right.$$

$$= \underline{\underline{2\pi i e^2}}$$

$$(8) \int_0^z e^{-t^2} dt$$

$$e^{-t^2} = 1 - \frac{t^2}{1!} + \frac{(t^2)^2}{2!} - \frac{(t^2)^3}{3!} + \dots$$

$$\int e^{-t^2} dt = t - \frac{t^3}{3(1!)} + \frac{t^5}{5(2!)} - \frac{t^7}{7(3!)} + \dots$$

$$\int_0^z e^{-t^2} dt = \underline{\underline{z - \frac{z^3}{3} + \frac{z^5}{10} - \dots}}$$

$$(9) z^{-5} \sin z = \frac{\sin z}{z^5}$$

$$= \frac{1}{z^5} \left[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots \right]$$

(4)

$$(10) \quad f(z) = \frac{-8}{1+z^2}$$

Singularities, $z = \pm i$ are simple poles.

$$\text{Res}_{z=i} f(z) = \lim_{z \rightarrow i} (z-i) \left(\frac{-8}{1+z^2} \right) = \lim_{z \rightarrow i} \frac{-8}{z+i} = \underline{\underline{-4i}}$$

$$\text{Res}_{z=-i} f(z) = \lim_{z \rightarrow -i} (z+i) \left(\frac{-8}{1+z^2} \right) = \lim_{z \rightarrow -i} \frac{-8}{z-i} = \underline{\underline{4i}}$$

PART B

11(a) Lagrange's Equation is $Pp + Qq = R$
 Subsidiary equations are

$$\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz}$$

$$\begin{aligned} \frac{dy}{2xy} &= \frac{dz}{2xz} \Rightarrow \frac{dy}{y} = \frac{dz}{z} \\ &\Rightarrow \log(y/z) = \log c \\ &\Rightarrow \boxed{\frac{y}{z} = c_1} \end{aligned}$$

⑤

Taking multipliers as x, y, z

$$\frac{x dx}{x^3 - xy^2 - xz^2} = \frac{y dy}{2xy^2} = \frac{z dz}{2xz^2}$$

$$\text{Adding, } \frac{x dx + y dy + z dz}{x^3 + xy^2 + xz^2} = \frac{dy}{2xy}$$

$$\Rightarrow \frac{x dx + y dy + z dz}{x^2 + y^2 + z^2} = \frac{dy}{2y}$$

$$\Rightarrow 2 \left(\frac{x dx + y dy + z dz}{x^2 + y^2 + z^2} \right) = \frac{dy}{y}$$

$$\text{Int} \Rightarrow \log(x^2 + y^2 + z^2) = \log y + \log b.$$

$$\Rightarrow \frac{x^2 + y^2 + z^2}{y} = c_2.$$

$$\text{General solution is } \phi\left(\frac{y}{x}, \frac{x^2 + y^2 + z^2}{y}\right) = 0$$

$$(b) \quad f(x, y, z, p, q) = (p^2 + q^2)y - qz = 0 \rightarrow \textcircled{1}$$

Charpit's subsidiary equations are

$$\frac{dx}{-2py} = \frac{dy}{z-2q} = \frac{dz}{-qz} = \frac{dp}{-pq} = \frac{dq}{p^2}$$

(6)

$$\text{Take } \frac{dp}{-pq} = \frac{dq}{p^2}$$

$$\text{Integrating, } \boxed{p^2 + q^2 = a}$$

$$\textcircled{1} \Rightarrow ay - qz = 0 \Rightarrow q = \frac{z}{ay}$$

$$\text{Now } (p^2 + q^2)y = qz \Rightarrow \boxed{q = \frac{c^2 y}{z}} \text{ and}$$

$$\boxed{p = \frac{c\sqrt{z^2 - c^2 y^2}}{z}}$$

$$\text{Now } dz = p dx + q dy$$

$$\Rightarrow z = \int \frac{c\sqrt{z^2 - c^2 y^2}}{z} dx + \int \frac{c^2 y}{z} dy$$

$$\underline{\underline{z^2 = (a + cx)^2 + c^2 y^2}}$$

(12) Equation of the plane is $lx + my + nz = a$
where $l^2 + m^2 + n^2 = 1$

$$lx + my + \sqrt{1 - l^2 - m^2} z = a \sqrt{l^2 + m^2 + n^2}$$

Diff. partially w.r.t 'x' and 'y'

$$\boxed{l = -np} \quad , \quad \boxed{m = -nq}$$

(7)

$$\text{ie } lx + my + \sqrt{1-m^2-l^2} z = a \sqrt{l^2+m^2+n^2}$$

$$\Rightarrow -np x + (-nq) y + \sqrt{1-n^2q^2-n^2p^2} z = a \sqrt{n^2q^2+n^2p^2+n^2}$$

$$\Rightarrow \underline{\underline{z = px + qy + a \sqrt{p^2+q^2+1}}}$$

(b) $u(x, y) = X(x) Y(y)$

Substituting in $3 \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0$

$$\Rightarrow 3 \frac{X'}{X} = -2 \frac{Y'}{Y} = k$$

Take $\frac{X'}{X} = \frac{k}{3} \Rightarrow \boxed{X = C_1 e^{\frac{kx}{3}}}$

$$-2 \frac{Y'}{Y} = k$$

$$\Rightarrow \frac{Y'}{Y} = -\frac{k}{2} \Rightarrow \boxed{Y = C_2 e^{-\frac{ky}{2}}}$$

$$u = XY = C_1 C_2 e^{\frac{kx}{3}} e^{-\frac{ky}{2}}$$

Now $u(x, 0) = 4e^{-x} \Rightarrow \boxed{C_1 C_2 = 4, k = -3}$

$$\Rightarrow \underline{\underline{u = 4e^{-x+3yh}}}$$

(12) Wave equation is $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$

Boundary conditions are $\left. \begin{matrix} y(0, t) = 0 \\ y(l, t) = 0 \end{matrix} \right\} \rightarrow (1)$

Initial conditions are $\left(\frac{\partial y}{\partial t} \right)_{t=0} = 0 \rightarrow (2)$

We have

$$y(x, 0) = \begin{cases} \frac{3ax}{l}, & 0 < x < l/3 \\ \frac{3a(l-2x)}{l}, & l/3 \leq x < 2l/3 \\ \frac{3a(x-l)}{l}, & 2l/3 \leq x \leq l. \end{cases}$$

The most general solution of Wave equation is

$$y(x, t) = (A \cos px + B \sin px) (C \cos cpt + D \sin cpt) \rightarrow (3)$$

Applying (1) in (3), we have.

$$A = 0 \text{ and } p = \frac{n\pi}{l}.$$

By condition (2), $D = 0$.

Generalising

$$\therefore y(x, t) = \sum_1^\infty b_n \sin \frac{n\pi x}{l} \cos \frac{n\pi ct}{l} \text{ where } b_n = \text{BC.}$$

By the condition for $y(x, 0)$, we have

(9)

$$y(x,0) = \sum_1 b_n \sin \frac{n\pi x}{l} \quad \text{where}$$

$$b_n = \frac{2}{l} \left[\int_0^{l/3} \frac{3ax}{l} \sin \frac{n\pi x}{l} dx + \int_{l/3}^{2l/3} \frac{3a(l-2x)}{l} \sin \frac{n\pi x}{l} dx + \int_{2l/3}^l \frac{3a(x-l)}{l} \sin \frac{n\pi x}{l} dx \right]$$

$$= \frac{2}{l} \left\{ \left(\frac{3ax}{l} \right) \left(-\frac{\cos \left(\frac{n\pi x}{l} \right)}{\frac{n\pi}{l}} \right) - \left(\frac{3a}{l} \right) \left(-\frac{\sin \left(\frac{n\pi x}{l} \right)}{\left(\frac{n\pi}{l} \right)^2} \right) \right\}_0^{l/3} \\ + \left\{ \left(\frac{3a(l-2x)}{l} \right) \left(-\frac{\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - \left(\frac{3a}{l} (-2) \right) \left(-\frac{\sin \frac{n\pi x}{l}}{\left(\frac{n\pi}{l} \right)^2} \right) \right\}_{l/3}^{2l/3} \\ + \left\{ \left(\frac{3a(x-l)}{l} \right) \left(-\frac{\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - \left(\frac{3a}{l} \right) \left(-\frac{\sin \frac{n\pi x}{l}}{\left(\frac{n\pi}{l} \right)^2} \right) \right\}_{2l/3}^l$$

$$= \frac{18a}{n^2\pi^2} \sin \frac{n\pi}{3} (1 + (-1)^n)$$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} \frac{18a}{n^2\pi^2} \sin \frac{n\pi}{3} (1 + (-1)^n) \sin \frac{n\pi x}{l}$$

10

14 Heat equation is $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$.

$$u(x,t) = (C_1 \cos px + C_2 \sin px) e^{-c^2 p^2 t}.$$

$$\frac{\partial u}{\partial x} = (-C_1 p \sin px + C_2 p \cos px) e^{-c^2 p^2 t}.$$

$$\frac{\partial u}{\partial x}(0,t) = 0 \Rightarrow \boxed{C_2 = 0}.$$

$$\frac{\partial u}{\partial x}(l,t) = 0 \Rightarrow \boxed{p = \frac{n\pi}{l}}$$

$$u(x,t) = C_1 \cos \frac{n\pi x}{l} e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}.$$

Generalising,

$$u(x,t) = \sum_{n=1}^{\infty} b_n \cos \frac{n\pi x}{l} \cdot e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}.$$

$$\Rightarrow u(x,0) = \sum_{n=1}^{\infty} b_n \cos \frac{n\pi x}{l} = x.$$

$$\Rightarrow b_0 = \frac{2}{l} \int_0^l x dx = \frac{2}{l} \left[\frac{x^2}{2} \right]_0^l = \frac{l}{2}.$$

$$b_n = \frac{2}{l} \int_0^l x \cos \frac{n\pi x}{l} dx.$$

$$= \frac{2}{l} \left[(x) \left(\frac{\sin \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (1) \left(\frac{-\cos \left(\frac{n\pi x}{l} \right)}{\left(\frac{n\pi}{l} \right)^2} \right) \right]_0^l$$

$$= \frac{2l}{n^2 \pi^2} [\cos n\pi - 1]$$

(11)

$$\therefore u(x,t) = \frac{l}{2} + \sum_{n=0}^{\infty} \frac{2l}{n^2 \pi^2} (\cos n\pi - 1) \cos \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2 t}{l^2}}$$

(15)

$$u = e^{-\pi x} \cos ay$$

(16)

$$u_x = -\pi e^{-\pi x} \cos ay, \quad u_y = e^{-\pi x} (\sin ay) a$$

$$u_{xx} = \pi^2 e^{-\pi x} \cos ay, \quad u_{yy} = -a^2 e^{-\pi x} \cos ay$$

$$u_{xx} + u_{yy} = 0 \Rightarrow e^{-\pi x} (\pi^2 - a^2) \cos ay = 0$$

$$\Rightarrow a = \pm \pi$$

$$u = e^{-\pi x} \cos \pi y$$

$$\text{By CR, } u_x = v_y \Rightarrow v_y = -\pi e^{-\pi x} \cos \pi y$$

$$\text{Integrating, } v = \frac{-\pi e^{-\pi x} \sin \pi y}{\pi} = e^{-\pi x} \sin \pi y + \phi(x)$$

$$\Rightarrow \frac{\partial v}{\partial x} = \pi e^{-\pi x} \sin \pi y + \phi'(x)$$

$$\text{Comparing } \phi'(x) = 1 \Rightarrow \underline{\phi(x) = x}$$

$\therefore$ Harmonic Conjugate,

$$\underline{\underline{v = e^{-\pi x} \sin \pi y + C}}$$

(12)

$$(b) \operatorname{Re}\left(\frac{z^2}{|z|^2}\right) = f(z)$$

$$\lim_{z \rightarrow 0} \operatorname{Re} \frac{z^2}{|z|^2} = \lim_{z \rightarrow 0} \operatorname{Re} \frac{(x+iy)^2}{|x+iy|^2}$$

$$= \lim_{z \rightarrow 0} \frac{x^2 - y^2}{x^2 + y^2}$$

Consider the straight line path $y = mx$

$$\lim_{z \rightarrow 0} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{x^2 - m^2 x^2}{x^2 + m^2 x^2} = \frac{1 - m^2}{1 + m^2},$$

which is not unique.

Hence $f(z)$ is not continuous at $z = 0$

$$(16) (a) |x+iy - 1/2| = 1/2.$$

$$\Rightarrow x^2 + y^2 - x = 0$$

$$\text{Now } w = 1/2 \Rightarrow x = \frac{u}{u^2 + v^2}, y = \frac{-v}{u^2 + v^2}$$

$$\text{Now } |z - 1/2| \leq 1/2 \Rightarrow x^2 + y^2 - x \leq 0.$$

$$\Rightarrow \frac{u^2}{(u^2 + v^2)^2} + \frac{v^2}{(u^2 + v^2)^2} - \frac{u}{u^2 + v^2} \leq 0$$

$$\Rightarrow 1 - u \leq 0$$

$$\Rightarrow \underline{\underline{u \geq 1}}$$

(13)

⑥ $f(z) = |z|^2 = x^2 + y^2$
 $\Rightarrow u(x, y) = x^2 + y^2, \quad v(x, y) = 0$
 $\Rightarrow \frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial x} = 0$
 $\frac{\partial u}{\partial y} = 2y, \quad \frac{\partial v}{\partial y} = 0$

So CR equations are satisfied only at $z=0$ and partial derivatives are continuous. Hence it is differentiable only at $z=0$. Hence it is nowhere analytic.

(17)

① $\oint_C \frac{z^3 - 6}{2z - i} dz$, C is $|z|=1$

Singularity is $z = \frac{i}{2}$ which lies inside $|z|=1$.

By Cauchy's Integral formula,

$$\oint_C \frac{f(z) dz}{z - z_0} = 2\pi i f(z_0)$$

$$\begin{aligned} \oint_C \frac{z^3 - 6}{2(z - i/2)} dz &= \frac{1}{2} (2\pi i \times f(i/2)) \\ &= \pi i \times \left(\frac{i^3}{8} - 6 \right) \end{aligned}$$

(14)

$$= \pi i \left[-\frac{1}{8} - 6 \right]$$

$$= \underline{\underline{\frac{\pi}{8} - 6\pi i}}$$

$$\textcircled{b} \quad \frac{z+2}{1-z^2} = \frac{(z+2)}{(1-z)(1+z)} = \frac{A}{1-z} + \frac{B}{z+1}$$

$$\Rightarrow A = \frac{3}{2}, \quad B = \frac{1}{2}$$

$$\frac{z+2}{1-z^2} = \frac{3}{2(1-z)} + \frac{1}{2(z+1)}$$

$$= \frac{3}{2}(1-z)^{-1} + \frac{1}{2}(z+1)^{-1}$$

$$= \frac{3}{2} \left[1 + z + z^2 + \dots \right] + \frac{1}{2} \left[1 - z + z^2 - \dots \right]$$

$$= \underline{\underline{2 + z + 2z^2 + 3z^3 + \dots}}$$

(15)

(18)

(a) Singularity is $z = 1/2$ of order 4 which lie inside $|z| < 1$

By Cauchy's formula for derivatives

$$\oint_C \frac{f(z) dz}{(z-z_0)^{n+1}} = \frac{2\pi i}{n!} f^{(n)}(z_0).$$

$$\oint_C \frac{\sinh 2z}{(z-1/2)^4} = \frac{2\pi i}{3!} f'''(1/2)$$

$$= \frac{\pi i}{3} \times 8 \cosh 1$$

$$= \frac{8\pi i}{3} \cosh 1$$

$$f(z) = \sinh 2z$$

$$f' = 2 \cosh 2z$$

$$f'' = 4 \sinh 2z$$

$$f''' = 8 \cosh 2z$$

$$f'''(1/2) = 8 \cosh 1$$

$$(b) \frac{1}{1+z} = \frac{1}{1+z+i-i}, \quad z_0 = -i$$

$$= \frac{1}{1-i^0} \left[\frac{1}{1 + \frac{z+i}{1-i}} \right]$$

$$= \frac{1}{1-i^0} \left[1 + \left(\frac{z+i}{1-i} \right) \right]^{-1}$$

$$= \frac{1}{1-i^0} \left[1 - \left(\frac{z+i}{1-i} \right) + \left(\frac{z+i}{1-i} \right)^2 - \dots \right]$$

(16)

(19)

(a)

$$\frac{e^z}{(z-1)^2} = \frac{e^{z-1+1}}{(z-1)^2} = e^{z-1} \cdot \frac{e}{(z-1)^2}$$

$$= e \left(\frac{e^{z-1}}{(z-1)^2} \right)$$

$$= \frac{e}{(z-1)^2} \left[1 + \frac{(z-1)}{1!} + \frac{(z-1)^2}{2!} + \dots \right]$$

Region is $<|z-1|<\infty$

(b)

$$z = i\theta, \quad d\theta = \frac{dz}{i^2 z}, \quad \sin \theta = \frac{z - 1/2}{2i}$$

$$\cos \theta = \frac{z + 1/2}{2}$$

$$0: 0 \rightarrow 2\pi \Rightarrow C \text{ is } |z|=1$$

$$\int_0^{2\pi} \frac{\sin^2 \theta}{5-4\cos \theta} d\theta = \oint_C \frac{\left(\frac{z-1/2}{2i}\right)^2 \frac{dz}{i^2 z}}{5-4\left(\frac{z+1/2}{2}\right)}$$

$$= \frac{1}{4i} \oint_C \frac{z^4 - 2z^2 + 1}{z^2(2z^2 - 5z + 2)} dz$$

Singularities are $z=0, 2, 1/2$ in which
0 and $1/2$ lie inside $|z|=1$

(17)

$z=0$ is a pole of order 2.

$$\therefore \text{Res } f(z) = \frac{1}{1!} \lim_{z \rightarrow 0} \frac{d}{dz} [(z-0)^2 \cdot f(z)]$$

$$= \underline{\underline{5/4}}$$

$$\text{Res } f(z) = \lim_{z \rightarrow 1/2} (z - 1/2) f(z) \quad / \quad z = 1/2 \text{ is a simple pole}$$

$$= \underline{\underline{-3/4}}$$

By Cauchy's Residue Theorem,

$$\oint_C f(z) dz = 2\pi i \left[\text{Sum of residues inside } C \right]$$

$$\int_0^{2\pi} \frac{\sin^2 \theta d\theta}{5-4\cos\theta} = \frac{1}{4i} \left[2\pi i \left(\frac{5}{4} - \frac{3}{4} \right) \right]$$

$$= \underline{\underline{\frac{\pi}{4}}}$$

(18)

(20)

(a) $\oint_C \tan 2\pi z \cdot dz, \quad |z - 0.2| = 0.2$

$$f(z) = \frac{\sin 2\pi z}{\cos 2\pi z}$$

Singularities are given by $\cos 2\pi z = 0$

$$\Rightarrow z = \frac{2n+1}{4}, \quad n = 0, \pm 1, \pm 2, \dots$$

Here $z = 1/4$ lie inside $|z - 0.2| = 0.2$.

$$\text{Res } f(z)_{z=1/4} = \frac{P(1/4)}{Q'(1/4)} = \underline{\underline{-\frac{1}{2\pi}}}$$

By Residue Theorem, $\oint_C \tan 2\pi z \, dz = 2\pi i \left[-\frac{1}{2\pi} \right]$
 $= \underline{\underline{-i}}$

(b) $\int_{-\infty}^{\infty} \frac{dx}{(x^2 - 3x + 2)(x^2 + 1)} = \oint_C \frac{dz}{(z^2 - 3z + 2)(z^2 + 1)}$

Singularities are $z = 1, 2, \pm i$ in which $z = 1, 2, i$ lie in the upper plane and $-i$ in the lower plane.

$$\text{Res } f(z)_{z=1} = \lim_{z \rightarrow 1} (z-1)f(z) = -1/2 \quad \left| \text{Res } f(z)_{z=i} = \frac{3-i}{20} \right.$$

$$\text{Res } f(z)_{z=2} = \lim_{z \rightarrow 2} (z-2)f(z) = 1/5$$

$$\text{Principal Value} = 2\pi i \left[\frac{3-i}{20} \right] + \pi i \left[-\frac{1}{2} + \frac{1}{5} \right] = \underline{\underline{\frac{\pi}{10}}}$$